

Dexter Fernandes

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Computer Vision Engineer

Computer Vision Engineer with 3+ years building and deploying vision systems for industrial manufacturing. Most of that work has focused on quality inspection and defect detection running at the edge under production constraints, including a YOLOv5 pipeline for paintjob defect detection on John Deere conveyor lines. Experienced across the full model lifecycle, with particular depth in the less visible parts: dataset curation, convergence debugging, failure-mode analysis, and keeping models reliable on noisy real-world camera feeds.

WORK EXPERIENCE

Esbaar • Muscat, Oman

Jan 2023 - Aug 2023

AI Engineer

- Owned the end-to-end computer vision pipeline for real-time perception systems in harsh industrial environments, from data curation and model training through deployment and on-site validation, improving F1 by 15% while maintaining deterministic 24/7 operation.
- Debugged training dynamics and convergence issues in production CV models, balancing model complexity, augmentation, and validation strategy to improve generalisation under lighting variation, occlusion, motion artefacts, and camera contamination.
- Built multimodal data ingestion, cleansing, training, and validation pipelines across image, video, and sensor data using Python, Pandas, Airflow, MLflow, DVC, Optuna, and CVAT, cutting model iteration time by 40% and reducing rollout time from 3 days to under 24 hours.
- Applied image preprocessing and artefact-detection workflows with OpenCV and scikit-image to identify mud and sand contamination on camera feeds, reducing false positives by 13% in harsh industrial environments.
- Hardened deployed vision systems with Dockerised microservices, Kafka messaging, structured logging, health checks, retries, and safe restarts, reducing live video pipeline crashes by 67% and improving root-cause analysis.
- Built benchmarking and profiling harnesses to evaluate open-source models (YOLOv5, TensorFlow Model Zoo) and quantify accuracy, latency, and stability pre and post optimisation for regression detection.

LivNSense Technologies • Bengaluru, India

May 2020 - Dec 2022

Lead Software Engineer - AI

- Built YOLOv5-based visual quality inspection for John Deere engine paintjobs on a conveyor line, detecting paint defects, contamination, and colour inconsistency as part of a manufacturing anomaly detection workflow.
- Led a team to design, train, and deploy production object detection and tracking models for industrial safety applications, reducing false positives by 33% while meeting latency and power constraints.
- Built low-latency perception pipelines using OpenVINO, and C++17, increasing throughput from 5 FPS to 15 FPS and delivering deterministic real-time inference on Intel VPU hardware under tight power and memory constraints.
- Used confusion-matrix-driven failure clustering to diagnose recurring CV failure modes, prioritise dataset fixes, and improve model robustness across camera feeds.
- Built automated labelling, dataset versioning, and performance-tracking workflows, repurposing existing detection models for annotation and cutting manual labelling effort by 80% while accelerating iteration cycles on large real-world datasets.
- Developed adversarial-condition training and validation conditions simulating occlusion, lighting variation, and motion artefacts, cutting performance variance by 25% and increasing resilience.
- Developed synthetic data generation and augmentation tools to improve edge-case coverage and cut manual labelling time by 60% using OpenCV and Albumentations.

- Built and maintained large-scale image and sensor datasets, automating labelling, versioning, and performance tracking to support anomaly investigation and continuous improvement cycles.

PROJECTS

Modern Edge CV Benchmarking

Mar 2026

- Benchmarked modern computer vision models including YOLOv8, YOLOv10, RT-DETR, EfficientViT, Grounding DINO, MobileSAM, and Depth Anything v2 to compare accuracy, latency, model size, and deployment suitability.
- Evaluated Vision Transformer and foundation-model approaches for object detection, segmentation, depth estimation, and open-vocabulary visual understanding, with a focus on production-style CV trade-offs.
- Exported models to ONNX and evaluated OpenVINO deployment paths, strengthening practical understanding of model optimisation, inference acceleration, and edge deployment constraints.
- Built benchmark comparisons using mAP, inference time, and model-size measurements to support model selection for low-latency visual AI systems.

End-to-End Damage Severity & Claim Assessment

Nov 2025

- Developed an automotive claims assessment pipeline combining computer vision-based damage detection with severity classification to support faster triage of real-world vehicle damage scenarios.
- Built a multimodal image-to-structured-output workflow that converted vehicle-damage detections and severity predictions into insurer-friendly claim summaries using structured reasoning and LLM-generated explanations.
- Designed the full workflow across data preprocessing, model training, evaluation, and a user-facing inference module, demonstrating end-to-end delivery from model output to usable product insight.
- Created interpretable repair insights from CV model outputs, translating visual predictions into clear claim explanations for non-technical users.

EDUCATION

Master of Science in Robotics - Merit

University of Bristol • Bristol, England

Sep 2023 - Sep 2024

Dissertation: Radiological SLAM with LiDAR Odometry

- Radiological SLAM with LiDAR Odometry. Designed and implemented a real-time SLAM system using RTAB-Map on the Clearpath Husky platform, integrating LiDAR and radiological data for localisation and mapping in inspection scenarios.
- Evaluated RTAB-Map, LIO-SAM, and Cartographer across comparable scenarios using ROS2 and rosbag2-based workflows to assess mapping and localisation behaviour.
- Validated sensor placement for coverage and calibrated LiDAR extrinsics to the robot base, including the Velodyne VLP-16 and Intel RealSense D435i.
- Implemented a ROS2 (rclcpp) point cloud pipeline in C++17 using PCL, and developed ROS2 nodes with tf2, message synchronisation, and sensor data playback workflows to maintain frame consistency and debug transform timing issues.
- Worked with camera and sensor ingestion in ROS2 pipelines, including timestamp alignment and downstream processing for SLAM and perception tasks.

Bachelor of Technology in Electrical and Electronics Engineering

Visvesvaraya National Institute of Technology • Nagpur, India

Jul 2016 - May 2020

- Built a strong foundation in signals, systems, embedded hardware, and control, which later supported work in real-time AI, sensor systems, and robotics software.

SKILLS

- **Computer Vision & Image Processing:** Anomaly detection, Data augmentation, Defect detection, Image processing, Object detection, Object tracking, OpenCV, PIL, scikit-image, Segmentation, Sensor calibration, Visual quality inspection

- **Deep Learning and Optimisation:** Bayesian Optimisation, Convolutional Neural Networks, Data Augmentation, FP16/INT8 deployment, Layer fusion, Object Detection & Tracking, ONNX, OpenVINO, Optuna, Post-training Quantisation, PyTorch, Quantisation-Aware Training, Scikit-Learn, Structured pruning, TensorFlow, TensorRT, YOLO
- **Datasets, Annotation & Evaluation:** Automated labelling, Cityscapes, COCO, Confusion-matrix-driven failure clustering, CVAT, DVC, F1-score, ImageNet, IoU, IoU, KITTI, Latency benchmarking, Latency benchmarking, mAP, model-size benchmarking, Model-size benchmarking, PASCAL VOC, Precision/Recall, Synthetic data generation
- **Cloud, Edge & MLOps:** Apache Airflow, Apache Kafka, API design, AWS, CI/CD, Docker, DVC (Data Version Control), Health checks, Intel VPUs, Microsoft Azure, MLFlow, Post-deployment validation, Production monitoring, Structured logging
- **Programming & Tooling:** Bash, C++17, CMake, Git, Linux, NumPy, Pandas, PostgreSQL, Python, Scikit-Learn
- **Computer Vision Models:** ByteTrack, DeepLabv3+, Depth Anything v2, EfficientDet, EfficientViT, Faster R-CNN, Grounding DINO, Mask R-CNN, MobileSAM, RT-DETR, SegFormer, SSD MobileNet V2, U-Net, YOLOv10, YOLOv5, YOLOv8
- **Edge AI and Inference:** Intel Core i7 (VTC7252-7C4IP), Intel Movidius Myriad VPU, NVIDIA Jetson AGX Xavier
- **Professional Skills:** Agile framework, Experiment-driven development, Stakeholder communication, Technical leadership
- **AI-Assisted Development:** ChatGPT Codex, Claude Code, Prompt engineering